

LESSON 9-3 • SESSION 1 • PRACTICE • FORM B

Equations from Tables

Learning target – I can write an equation that matches a table of values. • 6.EE.C.9

REMEMBER

- A **constant rate** means $y \div x$ is the same in every row. Then $y = (\text{rate}) \times x$.
- If $y - x$ is the same in every row, $y = x + (\text{that number})$.
- If neither works, see how much y grows each step: x : 1, 2, 3 and y : 5, 7, 9 gives $y = 2x + 3$.

WORKED EXAMPLE

Write an equation for the pairs (1, 6), (2, 12), (5, 30).

Step 1 $6 \div 1 = 6$, $12 \div 2 = 6$, $30 \div 5 = 6$.**Step 2** The rate is 6 every time.**Step 3** Check: $6 \times 5 = 30$.**Answer** $y = 6x$ **WORDS** equation (*ecuación*) • constant (*constante*) • rate (*tasa*)

PRACTICE

show your work • label units

1. Write an equation for each table.

a. x : 1, 2, 3 y : 6, 12, 18 $y =$ _____

b. x : 1, 2, 3 y : 11, 22, 33 $y =$ _____

c. x : 1, 2, 3 y : 8, 9, 10 $y =$ _____

d. x : 3, 6, 9 y : 12, 24, 36 $y =$ _____

- 2.
- Climbing Gym Passes.**
- Each pass costs the same. Write an equation for the cost
- y
- of
- x
- passes. Then find the cost of 8 passes.

Hint Divide a cost by its number of passes.

Passes (x)	1	2	3	5
Cost in \$ (y)	9	18	27	45

Equation; cost _____

- 3.
- Choir Risers.**
- Each riser holds the same number of singers. A table shows
- x
- : 1, 2, 3, 4 risers and
- y
- : 7, 14, 21, 28 singers. How many singers fit on 11 risers?

Hint Find the amount for 1 first.

Show your work.

Answer _____ singers

- 4.
- Water Bottles.**
- Each case holds the same number of bottles. A table shows
- x
- : 1, 2, 3 cases and
- y
- : 24, 48, 72 bottles. Write the equation. What does
- x
- stand for?

Show your work.

Equation _____

- 5.
- Notebook Order.**
- Each notebook costs the same. Write an equation. Then find the cost of 21 notebooks.

Hint Use one row to find the price of one.

Notebooks (x)	6	9	15
Cost in \$ (y)	24	36	60

Equation; cost _____

- 6. Mia's Rule.** A table shows x : 2, 4, 8 y : 16, 32, 64. Mia wrote $y = x + 14$. Show that it does not fit every row. Then write the correct equation.

Hint Test the rule on the second row.

Show your work.

Equation _____

- 7. Locker Stacks.** Every stack has the same number of lockers. Write an equation. Tell what x and y stand for.

Hint Find the lockers in one stack.

Stacks (x)	3	8	14
Lockers (y)	18	48	84

Equation _____

- 8. With a Constant.** A table shows x : 1, 2, 3 y : 7, 10, 13. $y \div x$ is not the same in every row. Write an equation. Then find y when $x = 10$.

Show your work.

Equation; y _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Micah's Work.** A table shows x : 3, 6, 9 and y : 24, 48, 72. Micah wrote the equation $x = 8y$. Explain his mistake and write the correct equation.

TEST PRACTICE

choose the best answer

- 10.** For x : 1, 2, 3 and y : 12, 24, 36, the equation is –

A $y = x + 11$

B $x = 12y$

C $y = 12x$

D $y = 13x$

- 11.** For x : 1, 2, 3 and y : 7, 9, 11, the equation is –

A $y = 2x + 5$

B $y = 7x$

C $y = x + 6$

D $y = 2x$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

$x = 9y$ Q	$y = 8x$ E	$y = 6x$ S	39 X	77 A	17 Z	84 U	$y = 24x$ L	$y = 9x$ V
#2	#3	#4	#5	#6	#7			

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____